Cambridge (CIE) A Level Maths: Pure 1

Your notes

Graphs of Functions

Contents

- Sketching Polynomials
- Sketching Reciprocal Graphs
- Solving Equations Graphically
- Proportional Relationships

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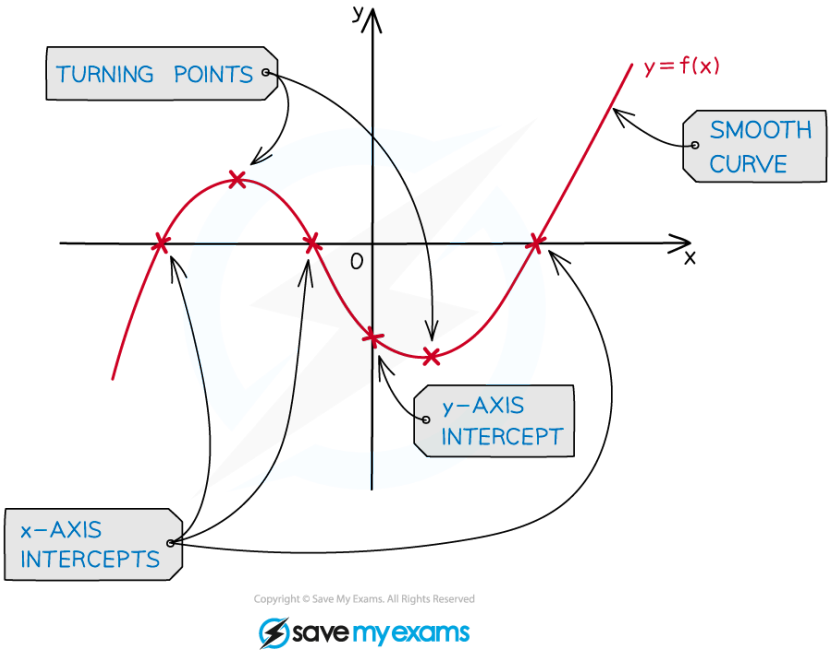
Your notes

Sketching polynomials

How do I sketch the graph of a polynomial?

- A polynomial is any finite function with non-negative indices, that could mean a quadratic, cubic, quartic or higher power



- When asked to sketch a polynomial you'll need to think about the following
  - **y**-axis intercept
  - **x**-axis intercepts (**roots**)
  - turning points (**maximum** and/or **minimum**)
  - a smooth **curve** (this takes practice!)

What steps should I take when sketching a polynomial?

STEP 1 Find the **y**-axis intercept by setting **x = 0**

STEP 2 Find the **x**-axis intercepts (roots) by setting **y = 0**

STEP 3 Consider the shape and "start"/"end" of the graph

eg. a **positive cubic** graph starts in third quadrant ("bottom left") and "ends" in first quadrant ("top right")

STEP 4 Consider where any turning points should go

STEP 5 Draw with a smooth curve

Your notes

$y = x^3 - 4x$

STEP 1: SET x=0 TO FIND y-AXIS INTERCEPT

AT x=0, y=0 (0,0)

STEP 2: SET y=0 TO FIND x-AXIS INTERCEPTS (ROOTS)

AT y=0, $x^3 - 4x = 0$

$x(x^2 - 4) = 0$

DIFFERENCE OF TWO SQUARES

$x(x+2)(x-2) = 0$

$x = 0,\ x = -2,\ x = 2$

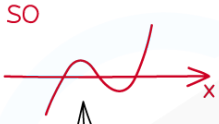
(0,0) (−2,0) (2,0)

ALSO THE y-AXIS INTERCEPT

STEP 3: CONSIDER SHAPE AND "START/END" OF GRAPH

$y = x^3 - 4x$ POSITIVE CUBIC

SO

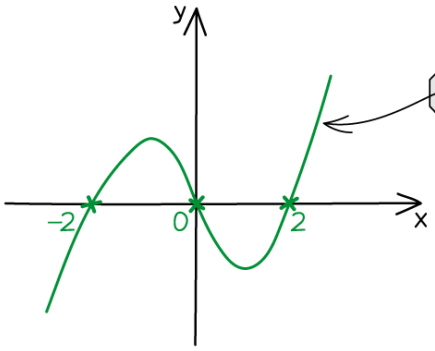


OR

ONLY THIS ONE WILL CROSS x-AXIS THREE TIMES

STEP 4: TURNING POINTS

- ONE MINIMUM
- ONE MAXIMUM



SMOOTH CURVE

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QUADRATICS ARE POLYNOMIALS (DEGREE 2)
ODD-DEGREE POLYNOMIALS **MUST** HAVE AT LEAST ONE x-AXIS INTERCEPT
EVEN-DEGREE POLYNOMIALS DO **NOT** NECESSARILY INTERCEPT THE x-AXIS

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- Coordinates of turning points can be found using **differentiation**
- Except with a point of inflection, repeated roots indicate the graph touches the **x**-axis

Worked Example

Your notes

Sketch the graph of $y = 3x^4 + 9x^3 + 6x^2$

$y = 3x^4 + 9x^3 + 6x^2$

STEP 1: y–AXIS INTERCEPT

AT $x = 0$, $y = 0$ $(0,0)$

STEP 2: x–AXIS INTERCEPT

AT $y = 0$ $3x^4 + 9x^3 + 6x^2 = 0$

$3x^2(x^2 + 3x + 2) = 0$

$3x^2(x+2)(x+1) = 0$

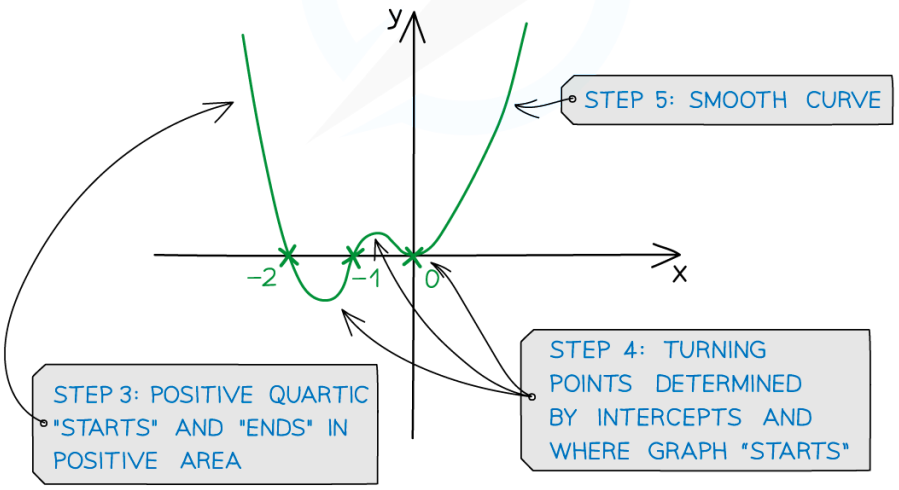
FULLY FACTORISED

$x = 0$ (REPEATED)

$x = -2$ $(-2,0)$

$x = -1$ $(-1,0)$

REPEATED, SO WILL TOUCH THE x–AXIS



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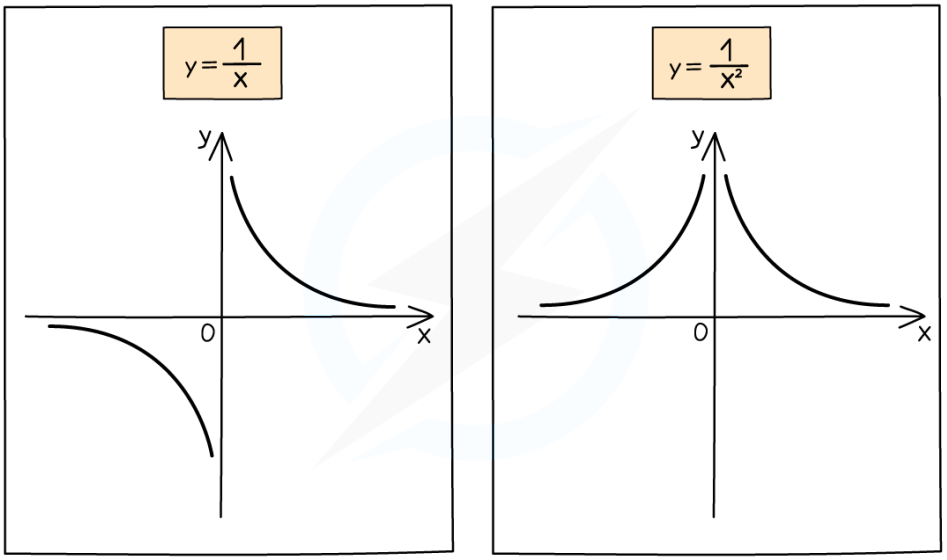
Sketching reciprocal curves

What are reciprocal graphs?

- Reciprocal graphs involve equations with an x term on the denominator e.g. $\frac{1}{x}$
- There are two basic reciprocal graphs to know for A level

$$y = \frac{1}{x} \text{ and } y = \frac{1}{x^2}$$

- The second one of these is always positive



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More reciprocal graphs

- You also need to recognise graphs where the numerator is not one

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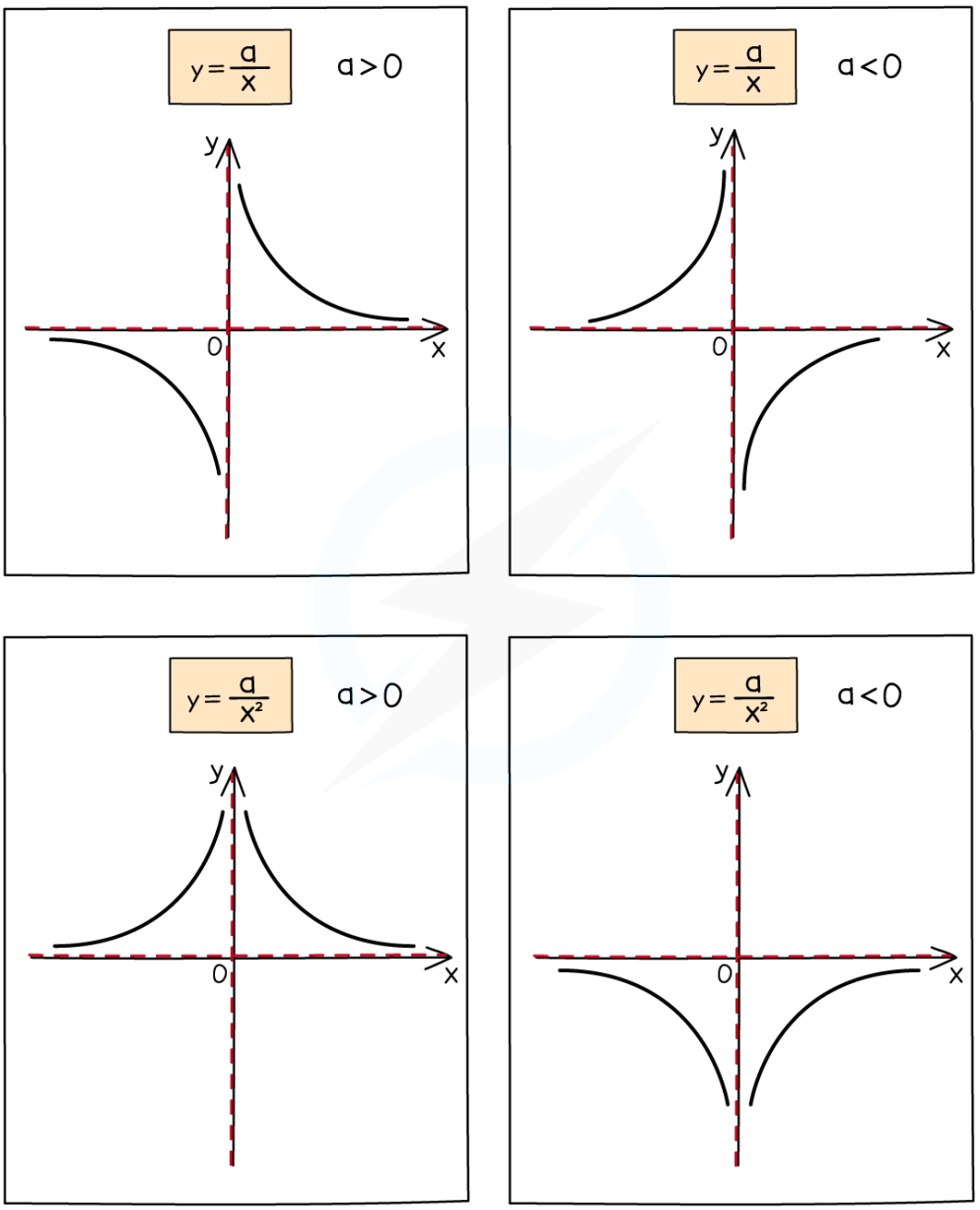

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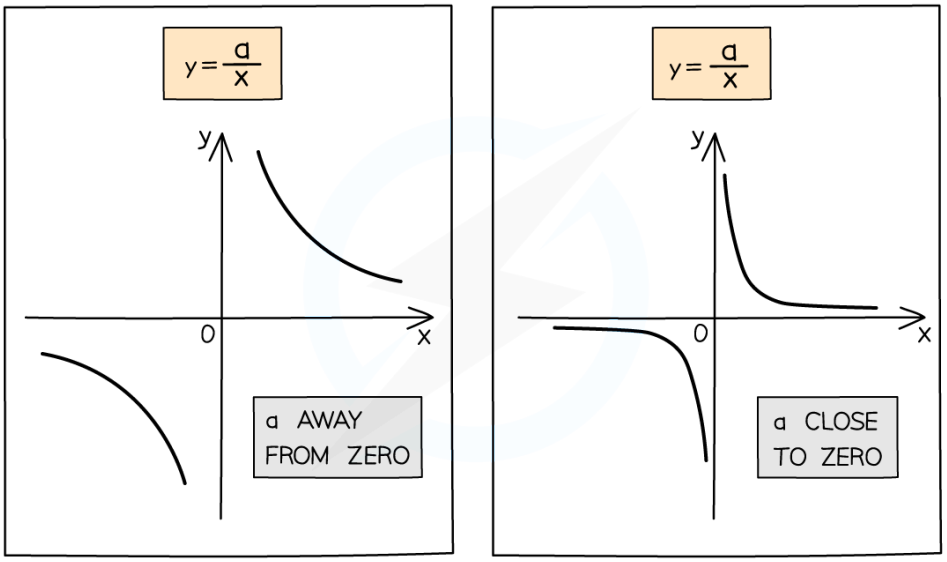
- The **sign** of **a** shows which part of the graph the curves are located
- The **size** of **a** shows how steep the curves are
  - The closer **a** is to 0 the more **L-shaped** the curves are



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- horizontal, **y = 0** (**x**-axis)
  - vertical, **x = 0** (**y**-axis)All have two asymptotes

How do I sketch a reciprocal graph?

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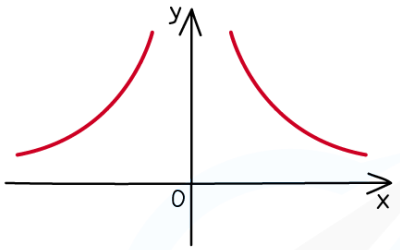
e.g. $y = \frac{6}{x^2}$

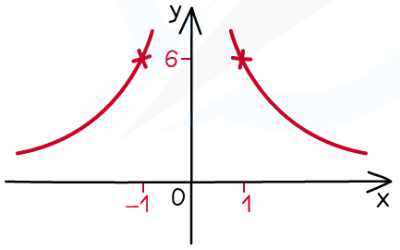
STEP 1: USE SIGN OF "a" TO LOCATE CURVES
USE SIZE OF "a" TO GAUGE STEEPNESS

a>0, CURVES IN POSITIVE AREAS
a=6, LARGE COMPARED TO 1, LESS L-SHAPED

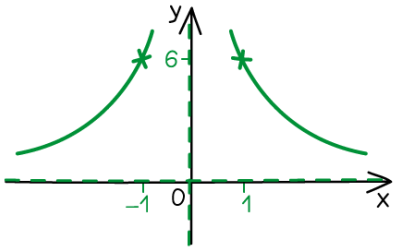
STEP 2: SKETCH THE GRAPH



STEP 3: LABEL x=1 & x=−1 AS A GUIDE



STEP 4: DRAW ASYMPTOTES WITH A DOTTED LINE



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STEP 1 Use the sign of "a" to locate the curves

and use the size of "a" to gauge the steepness of the curve

STEP 2 Sketch the graph

STEP 3 Label the points **x = 1** and **x = −1** as a guide to the scale of your graph

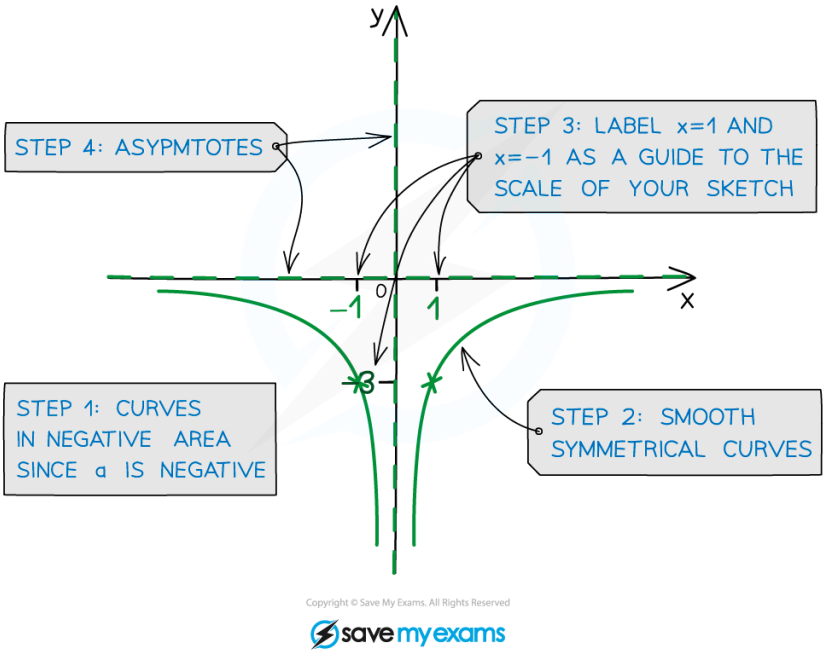
STEP 4 Draw asymptotes with a dotted line Draw **asymptotes** with **dotted** lines

- These graphs do not intercept either axis
- **Graph transformations** of them could cross the axes (see Translations)

Your notes

Worked Example

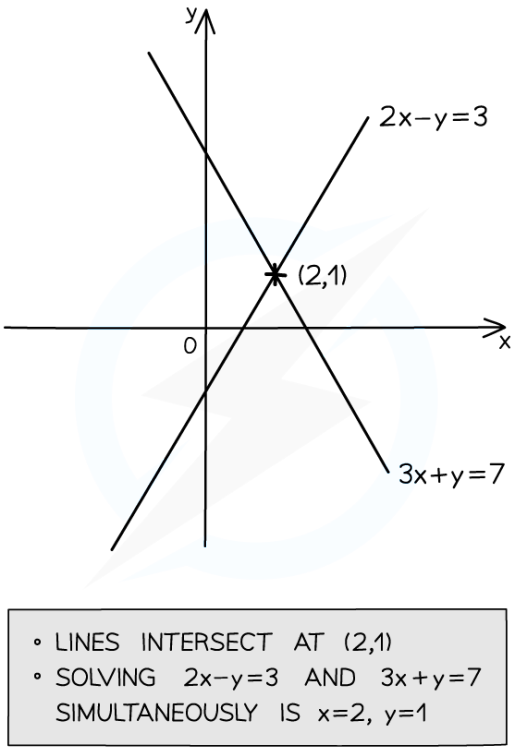
Sketch the graph of $y = \frac{-3}{x^2}$



Your notes

Solving equations graphically

What does solving simultaneous equations graphically mean?

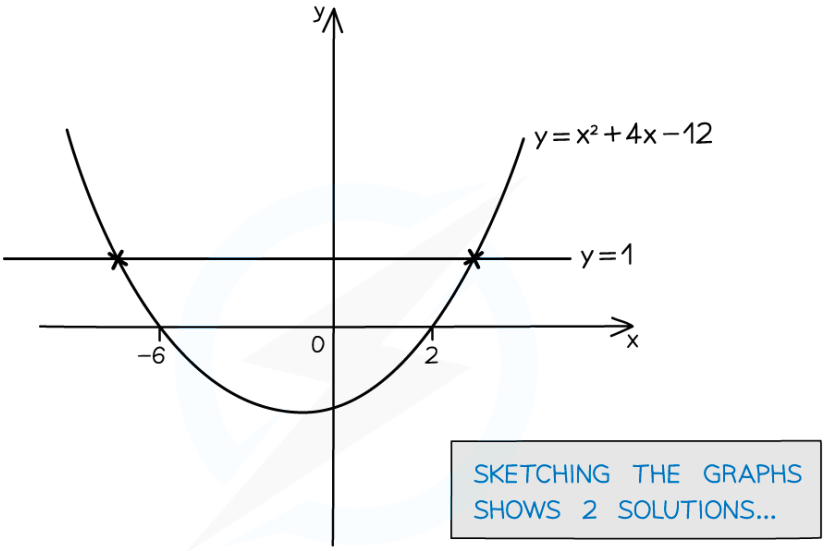


- This is a way to solve **simultaneous equations** (see Simultaneous Equations)
- Coordinates of the intersections are the **solutions**

How do I solve simultaneous equations using a graph?

Your notes

e.g. $y = x^2 + 4x - 12$
$y = 1$



$x^2 + 4x - 12 = 1$

$x^2 + 4x - 13 = 0$

$x = -2 + \sqrt{17}$ OR $x = -2 - \sqrt{17}$
$y = 1$ $\quad$ $y = 1$

... AND ALGEBRA FINDS THEM EXACTLY

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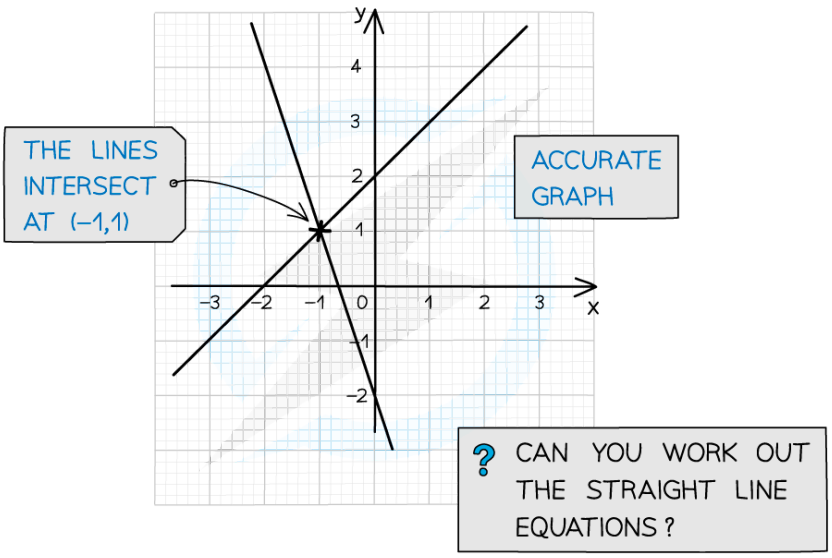
- Use **graphs** and **algebra** together
- **Sketch** a graph if it has not been given
- **Read/interpret** a graph if it has been given

Your notes



ANSWER: YES, $y = x + 2$, $y = -3x - 2$

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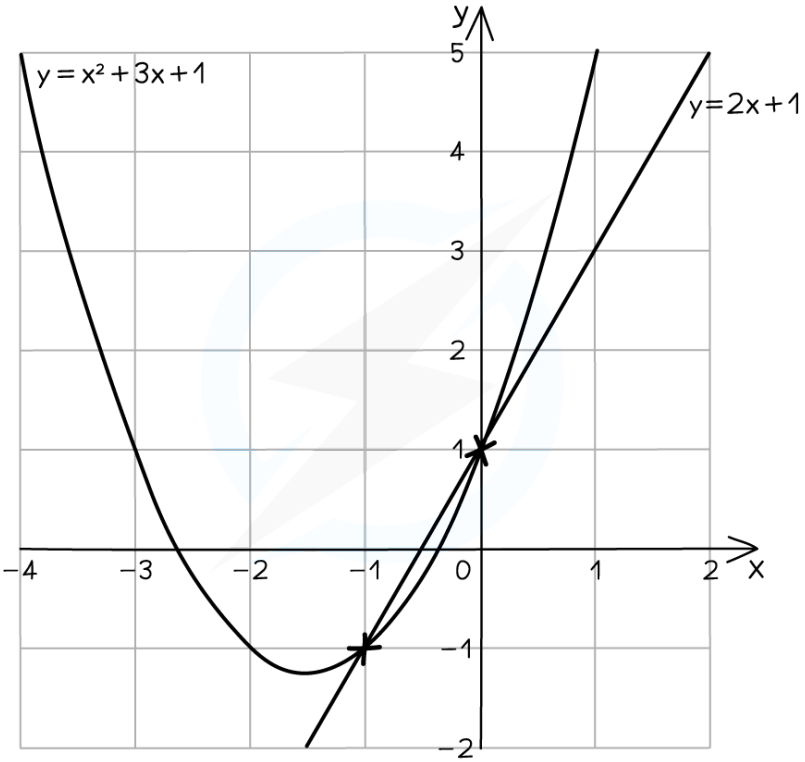
- It can be difficult to tell from a sketch if graphs intersect once, more, or not at all



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Your notes

Examiner Tips and Tricks

When writing out your solutions to simultaneous equations, always pair the correct x solution with the correct y solution

Worked Example

(a) Find all points of intersection between the curve with equation $y = 2x^3 + 5x^2 + 4$ and the straight line $2x + y = 4$

(b) Sketch the graphs on the same diagram clearly showing the points of intersection.

a) $y = 2x^3 + 5x^2 + 4$

$2x + y = 4$

THIS IS UNUSUAL BUT IT IS EASIER TO SUBSTITUTE THE CUBIC EQUATION INTO THE LINEAR ONE

$2x + 2x^3 + 5x^2 + 4 = 4$

$2x^3 + 5x^2 + 2x = 0$

$x(2x^2 + 5x + 2) = 0$

$x(2x+1)(x+2) = 0$

FACTORISE...

... AND SOLVE

$x = 0, \quad y = 4$

$x = -\frac{1}{2}, \quad y = 5$

$x = -2, \quad y = 8$

REMEMBER THE y-COORDINATES

$\therefore$ INTERSECTIONS ARE $(0,4)$, $(-\frac{1}{2},5)$ AND $(-2,8)$

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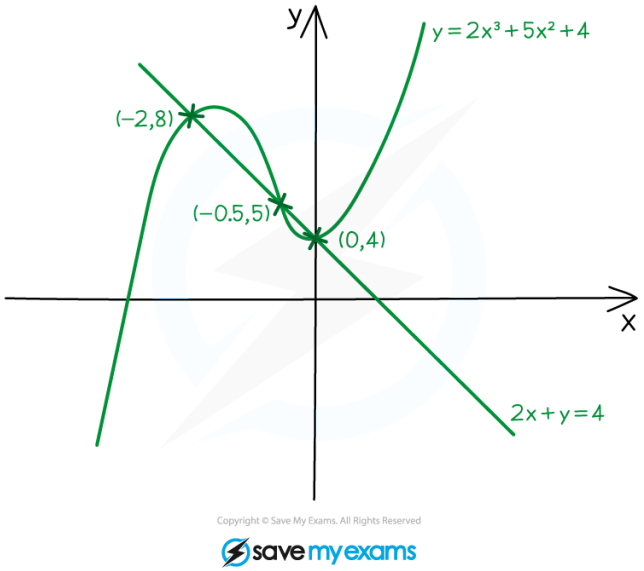
b)


y
x
$y = 2x^3 + 5x^2 + 4$
(−2,8)
(−0.5,5)
(0,4)
$2x + y = 4$
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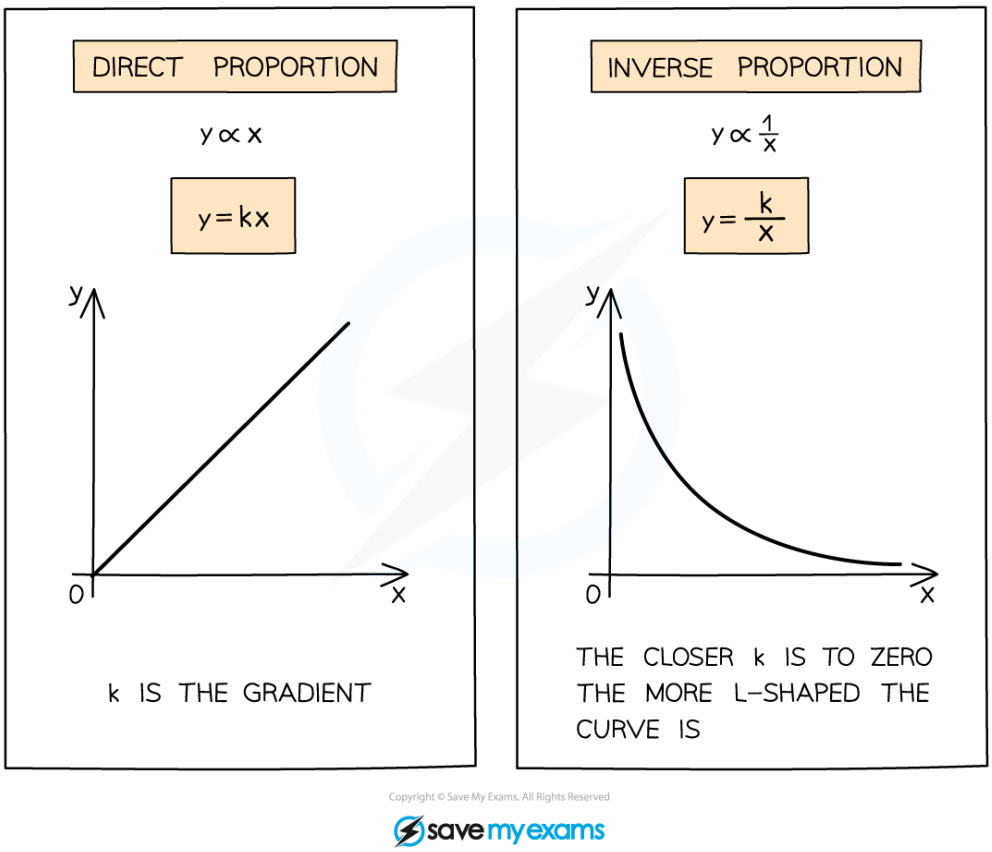
Your notes

Your notes

Proportional relationships

What are proportional relationships?

- Proportional relationships describe a proportional connection between two variables
- This can happen in two ways
  - Direct proportion $y = kx$
    - one variable increases or decreases the other does the same
  - Inverse proportion $y = \frac{k}{x}$
    - one variable increases the other decreases and vice versa
- Proportional relationships use the symbol $\propto$ which means is proportional to



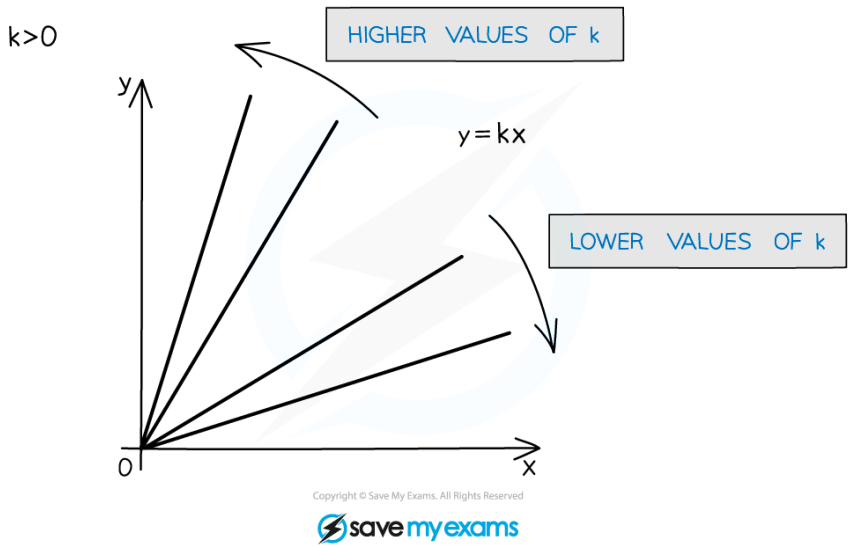
- Both direct and inverse proportion can be represented graphically

- Direct proportion creates a linear graph where k is the gradient
- Inverse proportion creates a reciprocal graph

Your notes

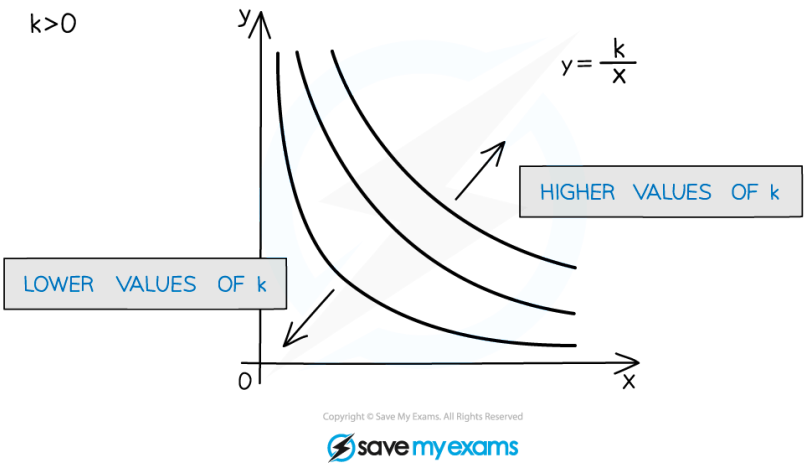
What is direct proportion?

- $y \propto x$ means y is proportional to x
- **y** increases as **x** does, **k** determines the **rate** (gradient)
- by changing this to the equation $y = kx$ we can substitute in given values and solve to find k
  - Note that this means the **ratio** of x and y is constant **k = y / x**



What is inverse proportion?

- $y \propto \frac{1}{x}$ means y is proportional to $\frac{1}{x}$ or y is inversely proportional to x
- **y** decreases as **x** increases and vice versa, **k** determines the rate
- by changing this to the equation $y = \frac{k}{x}$ we can substitute in given values and solve to find k
  - Note that this means the **product** of x and y is constant **k = xy**



- Set up your proportional relationship using $\propto$ then change to = k
- Be clear about what **y** is proportional to ...
  - "... the square of **x**" ($\mathbf{x^2}$)
  - "... **x** plus four" (**x + 4**)
- **Calculate** or **deduce** the value of **k** from the information given or a graph
- Once you've found k sub it back in to your original proportion equation
- You can now find any values using this proportional relationship
- **y = mx + c** rearranges to **y – c = mx** so (**y – c**) is **directly** proportional to **x**
- **Proportional relationships** are often used in modelling

Worked Example

Your notes

Your notes

(a) y is inversely proportional to the cube of x.

Given that $y = 3$ when $x = 2$ find the value of y when $x = 4$.

(b) Sketch the graph of y against x

a) $y \propto \frac{1}{x^3}$

INVERSE PROPORTION "... THE CUBE OF x"

$\therefore y = \frac{k}{x^3}$

AT (2,3): $3 = \frac{k}{2^3}$

USE THE GIVEN VALUES OF x AND y TO FIND k

$k = 3 \times 8$

$k = 24$

$\therefore y = \frac{24}{x^3}$

AT $x=4$, $y = \frac{24}{4^3}$

ONCE YOU KNOW THE EQUATION OF y YOU CAN SUBSTITUTE x=4 INTO IT

$y = \frac{3}{8}$

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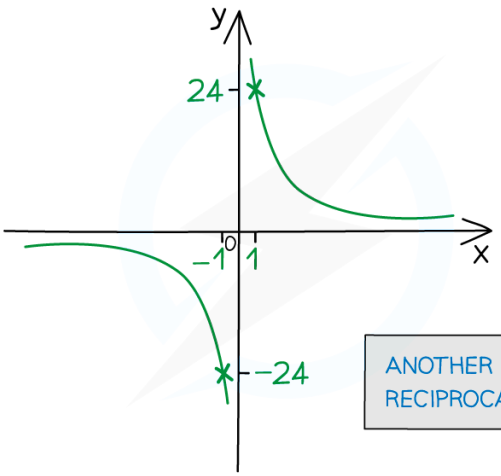
b)



ANOTHER TYPE OF RECIPROCAL GRAPH

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